

NCT Number: NCT00310180 (TAILORx)

Eligibility Criteria:

- Patients with operable histologically confirmed adenocarcinoma of the female breast who have completed primary surgical treatment and meet the following criteria:
- ER and/or progesterone receptor (PR)-positive: Estrogen and/or progesterone receptor positive disease (as defined by local pathology laboratory)
- Negative axillary nodes: As assessed by a sentinel lymph node biopsy, an axillary dissection, or both, and as defined by the Sixth Edition of the American Joint Committee on Cancer (AJCC) staging criteria
- Tumor size 1.1-5.0 cm (or 5 mm-1.0 cm plus unfavorable histological features):
- Unfavorable features defined as intermediate or poor nuclear and/or histologic grade, or lymphovascular invasion

Ineligibility Criteria:

- None found

NCT Number: NCT02819518 (KEYNOTE-355)

Eligibility Criteria:

- Has locally recurrent inoperable breast cancer not previously treated with chemotherapy and which cannot be treated with curative intent OR has metastatic breast cancer not previously treated with chemotherapy.
- Has centrally confirmed TNBC, as defined by the most recent American Society of Clinical Oncology/college of American Pathologists (ASCO/CAP) guidelines.
- Has completed treatment for Stage I-III breast cancer, if indicated, and ≥ 6 months elapsed between the completion of treatment with curative intent (e.g., date of primary breast tumor surgery or date of last adjuvant chemotherapy administration, whichever occurred last) and first documented local or distant disease recurrence.
- Has been treated with (neo)adjuvant anthracycline, if they received systemic treatment in the (neo)adjuvant setting, unless anthracycline was contraindicated or not considered the best treatment option for the participant in the opinion of the treating physician.
- Has measurable disease based on Response Evaluation Criteria in Solid Tumors version 1.1 (RECIST 1.1) as determined by local radiology review.
- Has provided recently or newly obtained core or excisional biopsy from a locally recurrent inoperable or metastatic tumor lesion for central determination of TNBC status and PD-L1 expression, unless contraindicated due to site inaccessibility and/or participant safety concerns.
- Has an Eastern Cooperative Oncology Group (ECOG) performance status of 0 or 1, as assessed within 10 days prior to the start of study drug.
- Has a life expectancy ≥ 12 weeks from randomization.
- Demonstrates adequate organ function, within 10 days prior to the start of study drug.
- Female participants are eligible to participate if they are not pregnant or breastfeeding AND they are not a woman of childbearing potential (WOCBP) OR is a WOCBP using a contraceptive method that is highly effective or is abstinent from heterosexual intercourse during the intervention period and for at least the time needed to eliminate each study intervention after the last dose of study intervention AND has a negative highly-sensitive pregnancy test ([urine or serum] as required by local regulations) within 24 hours (urine) or 72 hours (serum) before the first dose of study intervention.
- Male participants are eligible to participate if they agree to refrain from donating sperm during the intervention period and for at least the time needed to eliminate each study intervention after the last dose of study intervention PLUS be abstinent from heterosexual intercourse OR must agree to use contraception unless confirmed to be azoospermic.

Ineligibility Criteria:

- Is currently participating in a clinical study and receiving an investigational agent and/or using an investigational device, or has participated in a clinical study and received an investigational agent and/or used an investigational device within 4 weeks prior to randomization.
- Has not recovered (e.g., to $\leq$ Grade 1 or to baseline) from AEs due to a previously administered therapy.
- Has neuropathy $\geq$ Grade 2.
- Has an active autoimmune disease that has required systemic treatment in the past 2 years (e.g., with use of disease modifying agents, corticosteroids, or immunosuppressive drugs).
- Has a diagnosis of immunodeficiency or is receiving systemic steroid therapy or any other form of immunosuppressive therapy within 7 days prior to randomization.
- Has a known additional malignancy that progressed or required active treatment within the last 5 years. Exceptions include basal cell carcinoma of the skin, squamous cell carcinoma of the skin that has undergone potentially curative therapy, and in situ cervical cancer.
- Has known active central nervous system (CNS) metastases and/or carcinomatous meningitis. Participants with previously treated brain metastases may participate provided they have stable brain metastases and did not receive chemotherapy for metastatic breast cancer.
- Has history of (non-infectious) pneumonitis that required steroids or current pneumonitis.
- Has active, or a history of, interstitial lung disease.
- Has a known history of active tuberculosis (TB).
- Has an active infection requiring systemic therapy.
- Has a history of Class II-IV congestive heart failure or myocardial infarction within 6 months of randomization.
- Has a known psychiatric or substance abuse disorders that would interfere with cooperation with the requirements of the study.
- Is pregnant or breastfeeding, or expecting to conceive or father children within the projected duration of the study, starting with the screening visit through 120 days (or longer as specified by local institutional guidelines) after the last dose of study drug.
- Has received prior therapy with an anti-programmed cell death 1 (anti-PD-1), anti-PD-L1, or anti-PD-L2 agent or with an agent directed to another co-inhibitory T cell receptor (such as cytotoxic T-lymphocyte-associated protein 4 [CTLA-4], OX-40, CD137) or has previously participated in Merck pembrolizumab (MK-3475) clinical studies.
- Has a known history of human immunodeficiency virus (HIV).
- Has known active hepatitis B or hepatitis C.
- Has received a live vaccine within 30 days prior to randomization.

- Has a known history of hypersensitivity or allergy to pembrolizumab and any of its components and/or to any of the study chemotherapies (e.g., nab-paclitaxel, paclitaxel, gemcitabine, or carboplatin) and any of their components.
- Is receiving any medication prohibited in combination with study chemotherapies as described in the respective product labels, unless medication was stopped within 7 days prior to randomization.

NCT Number: NCT00567190 (CLEOPATRA)

Eligibility Criteria:

- Histologically or cytologically confirmed adenocarcinoma of the breast with locally recurrent or metastatic disease, and candidate for chemotherapy. Participants with measurable and non-measurable disease are eligible (locally recurrent disease must not be amenable to resection with curative intent; participants with de novo Stage IV disease are eligible)
- Human epidermal growth factor receptor 2 (HER2)-positive metastatic breast cancer (MBC)
- Left ventricular ejection fraction (LVEF) ≥ 50 percent (%) at baseline (within 42 days of randomization)
- Eastern Cooperative Oncology Group (ECOG) performance status (PS) 0 or 1
- For women of childbearing potential and men with partners of childbearing potential, agreement to use a highly effective form of contraception and to continue its use for the duration of study treatment and for at least 7 months after the last dose of study treatment

Ineligibility Criteria:

- History of anti-cancer therapy for MBC (with the exception of one prior hormonal regimen for MBC, which must be stopped prior to randomization)
- History of approved or investigative tyrosine kinase/HER inhibitors for breast cancer in any treatment setting, except trastuzumab used in the neoadjuvant or adjuvant setting
- History of systemic breast cancer treatment in the neo-adjuvant or adjuvant setting with a disease-free interval from completion of the systemic treatment (excluding hormonal therapy) to metastatic diagnosis of less than ($<$)12 months
- History of persistent Grade ≥ 2 hematologic toxicity resulting from previous adjuvant therapy
- Current peripheral neuropathy of National Cancer Institute-Common Terminology Criteria for Adverse Events (NCI-CTCAE) Version 3.0, Grade ≥ 3 at randomization
- History of other malignancy within the last 5 years, except for carcinoma in situ of the cervix, basal cell carcinoma or squamous cell carcinoma of the skin that has been previously treated with curative intent
- Current clinical or radiographic evidence of central nervous system (CNS) metastases
- Computed tomography (CT) or magnetic resonance imaging (MRI) scan of the brain is mandatory in cases of clinical suspicion of brain metastases
- History of exposure to cumulative doses of anthracyclines
- Current uncontrolled hypertension or unstable angina
- History of congestive heart failure (CHF) of any New York Heart Association (NYHA) criteria, or serious cardiac arrhythmia requiring treatment (exception: atrial fibrillation or paroxysmal supraventricular tachycardia)

- History of myocardial infarction within 6 months of randomization
- History of LVEF decline to below 50% during or after prior trastuzumab neo-adjuvant or adjuvant therapy
- Current dyspnea at rest due to complications of advanced malignancy, or other diseases that require continuous oxygen therapy
- Inadequate organ function, as defined in the protocol, within 28 days prior to randomization
- Current severe, uncontrolled systemic disease
- Major surgical procedure or significant traumatic injury within 28 days prior to study treatment start or anticipation of the need for major surgery during the course of study treatment
- Pregnant or lactating women
- History of receiving any investigational treatment within 28 days of randomization
- Current known infection with human immunodeficiency virus (HIV), hepatitis B virus (HBV), or hepatitis C virus (HCV)
- Receipt of IV antibiotics for infection within 14 days of randomization
- Current chronic daily treatment with corticosteroids (excluding inhaled steroids)
- Known hypersensitivity to any of the study drugs
- Assessed by the investigator to be unable or unwilling to comply with the requirements of the protocol

NCT Number: NCT01945775 (EMBRACA)

Eligibility Criteria:

- Histologically or cytologically confirmed carcinoma of the breast
- Locally advanced breast cancer that is not amenable to curative radiation or surgical cure and/or metastatic disease appropriate for systemic single cytotoxic chemotherapy
- Documentation of a deleterious, suspected deleterious, or pathogenic germline BRCA1 or BRCA2 mutation from Myriad Genetics or other laboratory approved by the Sponsor
- No more than 3 prior chemotherapy-inclusive regimens for locally advanced and/or metastatic disease (no limit on prior hormonal therapies or targeted anticancer therapies such as mechanistic target of rapamycin (mTOR) or CDK4/6 inhibitors, immune-oncology agents, tyrosine kinase inhibitors, or monoclonal antibodies against CTL4 or VEGF)
- Prior treatment with a taxane and/or anthracycline in the neoadjuvant, adjuvant, locally advanced, or metastatic setting unless medically contraindicated
- Have measurable or non-measurable, evaluable disease by the revised response evaluation criteria in solid tumors (RECIST) v.1.1
- Eastern Cooperative Oncology Group (ECOG) performance status ≤ 2

Ineligibility Criteria:

- First-line locally advanced and/or metastatic breast cancer with no prior adjuvant chemotherapy unless the Investigator determines that one of the 4 cytotoxic chemotherapy agents in the control arm would otherwise be offered to the subject
- Prior treatment with a PARP inhibitor (not including iniparib)
- Not a candidate for treatment with at least 1 of the treatments of protocol-specific physician's choice (ie, capecitabine, eribulin, gemcitabine, vinorelbine)
- Subjects who had objective disease progression while receiving platinum chemotherapy administered for locally advanced or metastatic disease; subjects who received low-dose platinum therapy administered in combination with radiation therapy are not excluded
- Subjects who have received platinum in the adjuvant or neoadjuvant setting are eligible; however, subjects may not have relapsed within 6 months of the last dose of prior platinum therapy
- Cytotoxic chemotherapy within 14 days before randomization
- Radiation or anti-hormonal therapy or other targeted anticancer therapy within 14 days before randomization
- HER2 positive breast cancer
- Active inflammatory breast cancer

- CNS metastases
- Exception: Adequately treated brain metastases documented by baseline CT or MRI scan that has not progressed since previous scans and that does not require corticosteroids (except prednisone ≤ 5 mg/day or equivalent) for management of CNS symptoms. A repeat CT or MRI following the identification of CNS metastases (obtained at least 2 weeks after definitive therapy) must document adequately treated brain metastases.
- Subjects with leptomeningeal carcinomatosis are not permitted
- Prior malignancy except for any of the following:
 - Prior BRCA-associated cancer as long as there is no current evidence of the cancer
 - Carcinoma in situ or non-melanoma skin cancer
 - A cancer diagnosed and definitively treated ≥ 5 years before randomization with no subsequent evidence of recurrence
- Known to be human immunodeficiency virus positive
- Known active hepatitis C virus, or known active hepatitis B virus
- Known hypersensitivity to any of the components of talazoparib

NCT Number: NCT01358877 (APHINITY)

Eligibility Criteria:

- Non-metastatic operable primary invasive HER2-positive carcinoma of the breast that is histologically confirmed, and adequately excised
- Eastern Cooperative Oncology Group (ECOG) performance status less than or equal to ($\leq$) 1
- Known hormone receptor status (estrogen receptor and progesterone receptor)
- The interval between definitive surgery for breast cancer and the first dose of chemotherapy must be no more than 8 weeks (56 days). The first cycle of chemotherapy must be administered within 7 days of randomization or on Day 56, whichever occurs first
- Baseline left ventricular ejection fraction (LVEF) greater than or equal to ($\geq$) 55 percent (%) measured by echocardiogram (ECHO) or Multiple-Gated Acquisition (MUGA) Scan
- Confirmed HER2 positive status
- Completion of all necessary baseline laboratory and radiologic investigations prior to randomization
- Women of childbearing potential and male participants with partners of childbearing potential must agree to use effective contraception (as defined by the protocol) by the participant and/or partner for the duration of the study treatment and for at least 7 months after the last dose of study drug

Ineligibility Criteria:

- History of any prior (ipsi- and/or contralateral) invasive breast cancer
- History of non-breast malignancies within the 5 years prior to study entry, except for carcinoma in situ of the cervix, carcinoma in situ of the colon, melanoma in situ, and basal cell and squamous cell carcinomas of the skin
- Any "clinical" T4 tumor as defined by primary tumor/regional lymph nodes/distant metastasis (TNM), including inflammatory breast cancer
- Any node-negative tumor
- Any previous systemic chemotherapy for cancer or radiotherapy for cancer
- Prior use of anti-HER2 therapy for any reason or other prior biologic or immunotherapy for cancer
- Concurrent anti-cancer treatment in another investigational trial
- Serious cardiac or cardiovascular disease or condition
- Other concurrent serious diseases that may interfere with planned treatment including severe pulmonary conditions/illness
- Abnormal laboratory tests immediately prior to randomization
- Pregnant or lactating women
- Sensitivity to any of the study medications or any of the ingredients or excipients of these medications

NCT Number: NCT01772472 (KATHERINE)

Eligibility Criteria:

- Adult patient, ≥ 18 years of age
- HER2-positive breast cancer
- Histologically confirmed invasive breast carcinoma
- Clinical stage T1-4/N0-3/M0 at presentation (patients with T1a/bN0 tumors will not be eligible)
- Completion of preoperative systemic chemotherapy and HER2-directed treatment consisting of at least 6 cycles of chemotherapy with a total duration of at least 16 weeks, including at least 9 weeks of trastuzumab and at least 9 weeks of taxane-based therapy
- Adequate excision: surgical removal of all clinically evident disease in the breast and lymph nodes as specified in protocol
- Pathological evidence of residual invasive carcinoma in the breast or axillary lymph nodes following completion of preoperative therapy
- An interval of no more than 12 weeks between the date of surgery and the date of randomization
- Known hormone-receptor status
- Eastern Cooperative Oncology Group (ECOG) performance status 0 or 1
- Adequate hematologic, renal and liver function
- Screening Left ventricular ejection fraction (LVEF) $\geq 50\%$ on echocardiogram (ECHO) or multiple-gated acquisition (MUGA) after receiving neoadjuvant chemotherapy and no decrease in LVEF by more than 15% absolute points from the pre-chemotherapy LVEF. Or, if pre-chemotherapy LVEF was not assessed, the screening LVEF must be $\geq 55\%$ after completion of neoadjuvant chemotherapy.
- For women who are not postmenopausal or surgically sterile: agreement to remain abstinent or use single or combined contraceptive methods that result in a failure rate of $< 1\%$ per year during the treatment period and for at least 7 months after the last dose of study drug
- Documentation of hepatitis B virus and hepatitis C virus serology is required

Ineligibility Criteria:

- Stage IV (metastatic) breast cancer
- History of any prior (ipsi- or contralateral breast cancer except lobular carcinoma in situ
- Evidence of clinically evident gross residual or recurrent disease following preoperative therapy and surgery
- Progressive disease during preoperative systemic therapy
- Treatment with any anti-cancer investigational drug within 28 days prior to commencing study treatment
- History of other malignancy within the last 5 years except for appropriately treated carcinoma in situ of the cervix, non-melanoma skin carcinoma, Stage I uterine cancer, or other non-breast malignancies with a similar

outcome to those mentioned above

- Patients for whom radiotherapy would be recommended for breast cancer treatment but for whom it is contraindicated because of medical reasons
- Current NCI CTCAE (Version 4.0) Grade ≥ 2 peripheral neuropathy
- History of exposure to the following cumulative doses of anthracyclines: Doxorubicin > 240 mg/m²; Epirubicin or Liposomal Doxorubicin-Hydrochloride (Myocet®) > 480 mg/m²; For other anthracyclines, exposure equivalent to doxorubicin > 240 mg/m²
- Cardiopulmonary dysfunction as defined by protocol
- Prior treatment with trastuzumab emtansine
- Current severe, uncontrolled systemic disease
- Pregnant or lactating women
- Any known active liver disease, e.g. due to HBV, HCV, autoimmune hepatic disorders, or sclerosing cholangitis
- Concurrent serious uncontrolled infections requiring treatment or known infection with HIV
- History of intolerance, including Grade 3 to 4 infusion reaction or hypersensitivity to trastuzumab or murine proteins or any components of the product

NCT Number: NCT02425891 (IMpassion130)

Eligibility Criteria:

- Metastatic or locally advanced, histologically documented TNBC characterized by absence of human epidermal growth factor 2 (HER2), estrogen receptor (ER), and progesterone receptor (PR) expression
- No prior chemotherapy or targeted systemic therapy for inoperable locally advanced or metastatic TNBC
- Eligible for taxane monotherapy (i.e., absence of rapid clinical progression, life-threatening visceral metastases, or the need for rapid symptom and/or disease control)
- A representative formalin-fixed, paraffin-embedded tumor specimen in paraffin blocks, or at least 20 unstained slides with an associated pathology report documenting ER, PR, and HER2 negativity. Participants with fewer than 20 unstained slides available at baseline, and not fewer than 12 unstained slides will be eligible upon discussion with Medical Monitor
- Eastern Cooperative Oncology Group performance status of 0 or 1
- Measurable disease as defined by RECIST v1.1
- Adequate hematologic and end-organ function

Ineligibility Criteria:

- Known central nervous system (CNS) disease, except for treated asymptomatic CNS metastases
- Leptomeningeal disease
- Pregnancy or lactation
- History of autoimmune disease
- Prior allogeneic stem cell or solid organ transplantation
- Positive test for human immunodeficiency virus
- Active hepatitis B or hepatitis C
- Receipt of a live, attenuated vaccine within 4 weeks prior to randomization, during treatment, or within 5 months following the last dose of atezolizumab/placebo

NCT Number: NCT02032823 (OLYMPIA)

Eligibility Criteria:

- Histologically confirmed non-metastatic primary invasive adenocarcinoma of the breast that is one of the following phenotypes:
 - Triple negative breast cancer defined as: ER and PgR negative AND HER2 negative (not eligible for anti-HER2 therapy)
 - ER and/or PgR positive, HER2 negative
- Documented germline mutation in BRCA1 or BRCA2 that is predicted to be deleterious or suspected deleterious (known or predicted to be detrimental/lead to loss of function).
- Completed adequate breast and axilla surgery.
- Completed at least 6 cycles neoadjuvant or adjuvant chemotherapy containing anthracyclines, taxanes or the combination of both. Prior platinum as potentially curative treatment for prior cancer (e.g. ovarian) or as adjuvant or neoadjuvant treatment for breast cancer is allowed.
- ECOG 0-1.

Ineligibility Criteria:

- Any previous treatment with a PARP inhibitor, including olaparib and/or known hypersensitivity to any of the excipients of study treatment.
- Patients with second primary malignancy. EXCEPTIONS are:
 - adequately treated non-melanoma skin cancer, curatively treated in situ cancer of the cervix, Ductal Carcinoma in situ (DCIS) of the breast, stage 1 grade 1 endometrial carcinoma
 - other solid tumours and lymphomas (without bone marrow involvement) diagnosed ≥ 5 years prior to randomisation and treated with no evidence of disease recurrence and for whom no more than one line of chemotherapy was applied.
- Concomitant use of known strong CYP3A inhibitors (e.g., itraconazole, telithromycin, clarithromycin, protease inhibitors boosted with ritonavir or cobicistat, indinavir, saquinavir, nelfinavir, boceprevir, telaprevir) or moderate CYP3A inhibitors (e.g., ciprofloxacin, erythromycin, diltiazem, fluconazole, verapamil). The required washout period prior to starting study treatment is 2 weeks. Concomitant use of known strong (e.g., phenobarbital, enzalutamide, phenytoin, rifampicin, rifabutin, rifapentine, carbamazepine, nevirapine and St John's Wort) or moderate CYP3A inducers (e.g., bosentan, efavirenz, modafinil). The required washout period prior to starting study treatment is 5 weeks for enzalutamide or phenobarbital and 3 weeks for other agents.
- Evidence of metastatic breast cancer

NCT Number: NCT00066690 (SOFT)

Eligibility Criteria:

- Premenopausal women (estradiol [E2] in the premenopausal range [according to institution parameters]) who meet the following criteria:
- Patients who did not receive chemotherapy should be randomized within 12 weeks after definitive surgery; such patients should have estradiol (E2) in the premenopausal range following surgery; the only patients who do not require testing of estradiol (E2) to confirm premenopausal status are those who have been menstruating regularly during the 6 months prior to randomization and have not used any form of hormonal contraception or any other hormonal treatments during the 6 months prior to randomization
- Patients who received prior adjuvant and/or neoadjuvant chemotherapy should be randomized after completing chemotherapy and within 8 months of the final dose of chemotherapy as soon as premenopausal status is confirmed; all such patients should have premenopausal status confirmed by an estradiol (E2) in the premenopausal range between 2 weeks and 8 months after completing chemotherapy
- Adjuvant trastuzumab (Herceptin ®) is allowable, and is not considered to be chemotherapy for eligibility timing determination

Ineligibility Criteria:

- None found

NCT Number: NCT00066703 (TEXT)

Eligibility Criteria:

- None found

Ineligibility Criteria:

- None found

NCT Number: NCT00545688 (NeoSphere)

Eligibility Criteria:

- female patients, ≥ 18 years of age;
- locally advanced, inflammatory or early stage invasive breast cancer;
- HER2 positive (HER2+++ by IHC or FISH/CISH+).

Ineligibility Criteria:

- metastatic disease (Stage IV) or bilateral breast cancer;
- previous anticancer therapy or radiotherapy for any malignancy;
- other malignancy, other than cancer in situ of the cervix, or basal cell cancer;
- insulin-dependent diabetes;
- clinically relevant cardiovascular disease.